

Media vs. Public Opinion on GLP-1 Weight Loss Drugs

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1 Introduction

GLP-1 receptor agonists such as Ozempic (semaglutide) and Wegovy are discussed simultaneously in patient communities and in mainstream media. Those two discourse spaces can shape different expectations about safety, efficacy, cost, and access. This study quantifies where those narratives align and where they diverge using a fully refreshed run of the project pipeline on the fixed 2024 collection window.

The corpus contains 1,286 public documents (1,184 Reddit posts and 102 WebMD reviews) and 647 media documents (647 news items). The analytical goal is to compare the two corpora along six dimensions: sentiment, lexical overlap, classifier separability, topic structure, aspect-level framing, and temporal coupling. In addition, a focused Reddit-vs-WebMD subanalysis is included to isolate differences within the public corpus.

Table 1: Run quality and representation checks (hybrid snapshot).

Metric	Value
Public documents	1,286
Media documents	647
Body coverage	0.696
Fallback ratio	0.304
Reddit / WebMD / News	1,184 / 102 / 647

2 Methodology

Collection strategy and data types. Public data were collected from Reddit (community posts) and WebMD (patient reviews), while media data were collected from Google News RSS with GLP-1-specific queries and optional article-body expansion. This yields mixed text types that reflect real consumption contexts: short conversational posts (Reddit), semi-structured patient narratives (WebMD), and publisher-formatted news coverage.

Information retrieval tools and rationale. Reddit collection used Arctic Shift endpoints; WebMD used deterministic HTML scraping; media collection used RSS retrieval, Google link decoding, and bounded HTML body extraction. This stack was selected because it is reproducible with open tools (`requests`, `BeautifulSoup`),

does not require proprietary access, and supports source traceability from raw collection to processed corpus.

Text analytics methods and rationale. Preprocessing used tokenization, lowercasing, stopword removal, and lemmatization to standardize cross-source language. VADER was used for sentiment because it is robust for short informal text and still interpretable on news prose. TF-IDF and Bag-of-Words were used for lexical comparison and features because they are transparent and suitable for domain terms. Supervised separability was tested with Multinomial Naive Bayes and KNN as complementary baselines (probabilistic linear vs distance-based). Topic structure used LDA and K-Means to triangulate latent themes and cluster separation. Aspect framing and temporal coupling were assessed to move beyond simple polarity into substantive narrative differences.

Experimentation process (decision path). The final configuration was chosen through explicit trial stages. First, we used snippet-only media text as a baseline for full corpus coverage and fast iteration. Second, we tested body-only media text to improve semantic depth, but coverage volatility made this representation brittle. Third, we compared `snippet`, `body`, and `hybrid` on the same frozen dataset and retained `hybrid` as the operating mode because it preserved body richness while controlling missing-body risk through fallback. For recollection, a strict run (timeout 15, min body tokens 80) failed representation gates, then a second run (timeout 25, min body tokens 60) passed and became the reporting snapshot.

Hyperparameter tuning and robustness checks. Classification used leakage-safe `Pipeline + StratifiedKfold + GridSearchCV`. Naive Bayes tuned α and KNN tuned neighborhood size k inside cross-validation. Robustness was tested in two ways: (1) length-normalized reruns (token cap = 40) to reduce document-length confounding, and (2) a within-public Reddit-vs-WebMD subanalysis to verify that public discourse is itself heterogeneous rather than a single homogeneous class.

3 Results

3.1 Sentiment Analysis

Public and media sentiment distributions differ significantly. Mean VADER polarity is 0.1117 for public discourse and 0.1905 for media discourse ($t = -2.301$, $p = 0.021516$, Cohen's $d = -0.108$). The effect size is small, but statistically detectable at corpus scale.

Interpretively, public content remains more mixed because personal journey narratives include both benefit reports and adverse-experience accounts, whereas media language is more likely to be templated around explanatory health reporting and policy narratives.

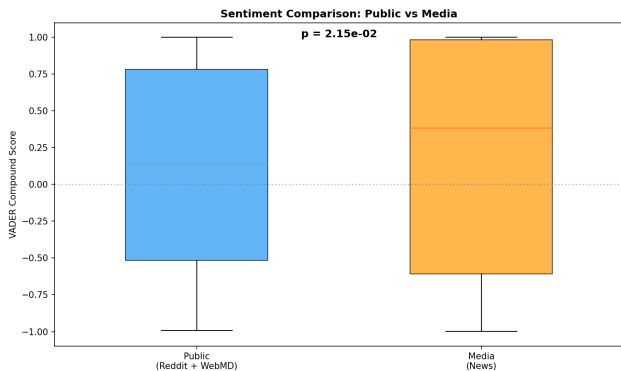


Figure 1: Sentiment comparison between public and media corpora (hybrid mode).

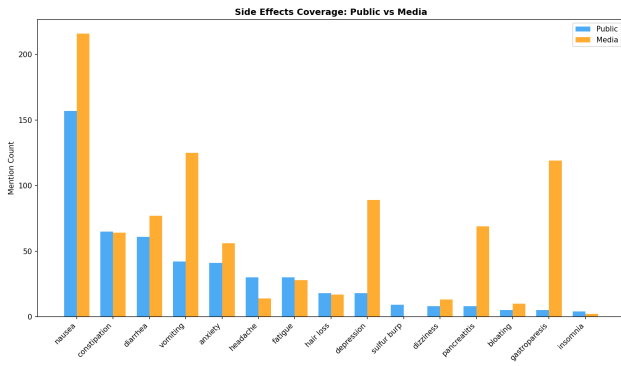


Figure 2: Side-effect mention counts in public versus media corpora.

3.2 Corpus Comparison

Lexical similarity is moderate at corpus level (cosine similarity = 0.446), and remains moderate under normalization (Norm40 cosine = 0.370). This pattern indicates shared topic anchors (drug names, efficacy, side effects) but distinct phrasing and contextual framing.

Side-effect framing differs materially in emphasis. For example, nausea prevalence in public documents is 0.0778 versus 0.0093 in media documents, showing that symptom reporting is more common in patient-authored text.

3.3 Text Classification

The two corpora are strongly separable in supervised classification. In leakage-safe 5-fold CV, Multinomial Naive Bayes reaches 0.9783 accuracy (best $\alpha = 0.1$) and KNN reaches 0.9550 (best $k = 5$). Under normalization, Naive Bayes remains 0.9757 and KNN improves to 0.9710, confirming that separability is not only a by-product of length differences.

These results indicate robust discourse-style divergence: public language is experiential and first-person, while media language is institutional, attribution-heavy, and policy-oriented.

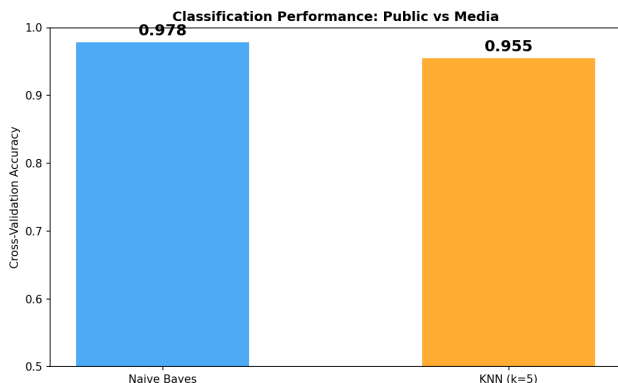


Figure 3: Cross-validated classification performance (NB vs KNN).

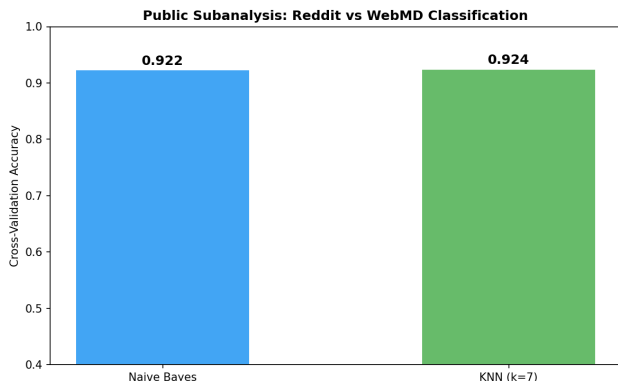


Figure 4: Public subanalysis: Reddit vs WebMD classification performance.

3.4 Public Subanalysis: Reddit vs WebMD

Within the public corpus, Reddit and WebMD remain distinct. Mean sentiment differs (Reddit 0.1479 vs WebMD -0.3077; $t = 6.885$, $p = 0.000000$, $d = 0.702$). Lexical overlap is moderate (cosine = 0.751), and source separability is high (NB = 0.9222, KNN = 0.9238 with $k = 7$). Reddit tends to emphasize terms such as weight, week, ozempic, dose, whereas WebMD emphasizes week, day, effect, side.

3.5 Topic Modeling

Topic modeling shows consistent thematic separation. Public topics are dominated by dosage progression, weight-change tracking, and day-to-day side-effect management. Media topics center on approvals, clinical evidence, shortages, and insurance-access framing. K-Means purity (0.898) supports this structural separation.

3.6 Aspect-Based Sentiment

Aspect-level sentiment reveals where framing differs most. Discrepancy scores (public minus media) are strongest for cost (-0.187), access (-0.132), and side effects (-0.174). Public discourse tends to encode these topics through direct personal experience, while media language is compar-

actively explanatory and policy-framed.

3.7 Temporal Analysis

Temporal coupling is weak. Cross-correlation peaks at lag -4 weeks ($r = -0.2763$), weekly volume correlation is 0.1894, and Granger tests are not significant in either direction (media->public $p = 0.76723$, public->media $p = 0.518258$). The two streams appear contemporaneously related but not strongly directional in this window.

4 Discussion

Three findings are stable in this refreshed hybrid run. First, discourse spaces remain clearly separable by language features, even after normalization-based robustness checks. Second, side-effect and access/cost framing are not symmetric across sources, with public posts carrying more direct adverse-experience emphasis. Third, weak temporal directionality suggests that weekly discourse changes are likely co-evolving rather than consistently led by one source.

A practical limitation is source heterogeneity: Reddit posts are shorter and conversational, WebMD reviews are structured but smaller in count, and news content varies by publisher formatting. Hybrid mode mitigates this by preferring full bodies while preserving coverage through controlled snippet fallback.

5 Conclusion

Using recollected data and hybrid media representation, this analysis confirms persistent narrative divergence between public discourse and media framing on GLP-1 drugs. The conclusions are grounded in explicit representation quality gates, reproducible end-to-end execution, and leakage-safe modeling, and they remain directionally consistent across multiple analytical methods.